

Welly AI Chat — Technical Documentation

n8n Workflow — Technical Overview (Redacted Version)

This is a redacted version of the internal technical documentation. All credentials, API keys, credential IDs, Airtable base/table IDs, phone number IDs, calendar account emails, and direct file links have been removed or replaced with generic placeholders. Structural and logical content — how the workflow behaves — has been preserved for reference.

1. Executive Summary & System Overview

The Welly AI Chat system is a fully automated WhatsApp customer service and sales bot built on n8n, deployed for a motorcycle dealership in Malaysia. The AI persona, "Jasmin," is a warm, multilingual Malaysian sales advisor that handles incoming WhatsApp messages, answers product FAQs, manages test ride bookings, and escalates complex inquiries to human staff. The system runs 24/7 without human intervention for routine queries.

1.1 Primary Functions

- Receive and process inbound WhatsApp messages from customers (polled on a short interval).
- Identify new vs. returning customers and retrieve conversation history.
- Classify the customer's intent using an LLM (FAQ query vs. booking action vs. general).
- Search relevant FAQ data from a database and inject it into the AI prompt.
- Generate a human-like, language-matched (English / Malay / Chinese) AI reply via an LLM.
- Handle test ride booking workflows: book, reschedule, cancel.
- Check calendar availability and enforce capacity limits (5 riders per session, 3 sessions per hour = 15 total).
- Create calendar events for confirmed bookings.
- Detect escalation triggers and route to a human staff member via a WhatsApp template message, using round-robin assignment.
- Log all conversations to the database (chat logs, customer records).
- Run a 1-hour follow-up wait on unanswered escalations and handle automated follow-ups.

1.2 Technology Stack

Component	Technology / Service	Purpose
Automation Engine	n8n (self-hosted)	Orchestrates the entire workflow
Messaging	WhatsApp (Cloud API)	Inbound & outbound messages
AI / LLM	Grok-4-Fast via OpenRouter	Natural language generation
Database	Airtable	Customer records, chat logs, FAQ, staff, escalations
Calendar	Google Calendar	Test ride slot management
API Gateway	OpenRouter	Routes LLM API calls
Credential Store	n8n credential vault	Stores API keys securely, referenced by ID

2. System Architecture & Data Flow

2.1 High-Level Architecture

The system operates as a polling-based pipeline rather than a pure webhook. A Schedule Trigger fires on a short interval and queries the database for any unprocessed incoming messages. Inbound WhatsApp messages are first written to a staging table, and n8n processes them asynchronously on each poll cycle.

2.2 Main Pipeline Stages

#	Stage	Description
1	Message Ingestion	Schedule trigger → search unprocessed record → mark as processed → normalize fields
2	Customer Lookup	Search customer table by phone number; create a new record if not found
3	History Retrieval	Search chat logs for last 6 exchanges; build a formatted chat-history string
4	Intent & FAQ	LLM classifies message category; fetch matching FAQ rows; combine into prompt context
5	Response Generation	Build final prompt; call the LLM; format reply text
6	Message Delivery	Send reply via WhatsApp; log to chat-log table
7	Post-Send Routing	Check reply for escalation trigger; route to staff or start 1-hour follow-up timer

2.3 Sub-Pipeline: Booking Actions

When the system detects a booking intent (book / reschedule / cancel), the main FAQ pipeline is bypassed and a dedicated booking sub-pipeline runs instead:

- Parse booking details deterministically (model, date, time, license).
- Use the LLM to validate and fill in missing required fields.
- Check calendar availability.
- Run capacity and duplicate checks (max 5 riders per session, 1 booking per rider).
- Create, update, or delete calendar events.
- Generate an LLM-crafted confirmation message with an indemnity-form link.
- Send a WhatsApp reply and log the exchange.

2.4 Database Architecture

Data is organised across three logical groupings, each holding a set of related tables:

Group	Table	Purpose & Key Fields
AI Logs	Raw Incoming	Stores raw inbound message data: message ID, From, To, Body, ProfileName, Processed flag, processed_at, Timestamp
AI Logs	Customers	Customer registry: CustomerID (phone number), Name (WhatsApp profile name)
AI Logs	Chatlogs	Full conversation history: CustomerID, Name, Intent, UserMessage, BotReply, Date, Timestamp
AI Logs	Escalation Logs	Records of escalations to human staff: staff ID, time, customer info
FAQ	General	FAQ knowledge base: Question, Answer, Remarks, Category. Remarks can flag escalation.
Staff Directory	Staff	Active staff for escalation: Name, ID (phone number), Phone, Status. Filter: Status $\neq$ "Inactive".

Specific base and table identifiers, along with all API credentials, are stored in the n8n credential vault and the internal configuration reference — not reproduced in this document.

3. Complete Node Reference (Summary)

The workflow contains 141 nodes organised into functional groups (A through K). Rather than reproducing the full node-by-node listing, this section summarises each functional group and its purpose. Detailed node configuration (exact field mappings, credential references, filter formulas) is documented in the internal n8n workflow file itself, which is the source of truth.

A: Message Ingestion Pipeline

Polls for new messages every 15 seconds, marks them as processed, and normalizes fields into a standard internal format (MessageSid, From, To, Body, ProfileName).

B: Customer Management

Looks up the customer by phone number; creates a new customer record on first contact.

C: Conversation History

Retrieves and formats the last 6 chat exchanges for the customer into a single context string.

D: Intent Classification & FAQ Retrieval

Uses the LLM to classify the message into a category (Pricing, Test Ride, Trade-In, Model Info, etc.), then retrieves matching FAQ entries from the knowledge base.

E: AI Response Generation (Main FAQ Flow)

Builds the full system prompt for the "Jasmin" persona (personality, language rules, escalation rules, dynamic context), then calls the LLM to generate the reply.

F: Message Delivery & Logging

Sends the AI-generated reply via WhatsApp and logs the exchange to the chat-log table.

G: Escalation to Human Staff

Detects escalation trigger phrases in the bot's reply, selects a staff member via round-robin, summarizes the conversation, and sends a WhatsApp template notification to staff.

H: Follow-Up Automation

After a 1-hour wait with no customer reply, checks whether a reply arrived; if not, may trigger further automated follow-up logic.

I: Booking Management Pipeline

Deterministically parses booking details (model, date, time, license) without an LLM, validates required fields with LLM assistance, checks calendar availability and capacity, and creates the booking.

J: Reschedule Pipeline

Mirrors the booking pipeline, but deletes the old calendar event and creates a new one once the new slot is confirmed available.

K: Cancellation Pipeline

Locates the customer's existing booking on the calendar (matched by phone number or name) and deletes it, then sends a cancellation confirmation.

4. Credentials, API Keys & Configuration

All credentials (LLM API access, database access, calendar OAuth2, WhatsApp API access) are stored in n8n's credential vault and referenced by name/ID inside the workflow — they are not hardcoded in individual nodes. Specific key values, credential IDs, base/table IDs, and phone number IDs are intentionally omitted from this document; refer to the secure internal credentials store for these values.

4.1 LLM API Details

Provider: OpenRouter. Model: Grok-4-Fast. Authentication: bearer token, managed via an n8n credential (not stored in node configuration).

Temperature settings by use case:

- FAQ category classification: 0.1 — highly deterministic.
- Booking field validation: effectively deterministic via JSON-only instruction.
- Main response generation (Jasmin's replies): 0.7 — creative but controlled.
- Chat summary for escalation: 0.5 — balanced.

4.2 WhatsApp Configuration

Inbound messages are received via the WhatsApp provider and staged into the database; the workflow polls this staging table on a short interval. Outbound messages are sent via the WhatsApp Cloud API using n8n's native WhatsApp node.

Staff escalation uses a pre-approved WhatsApp Business message template (name: "escalation", language: English) with 5 placeholder variables.

5. AI Persona: "Jasmin" — Prompt Logic Summary

"Jasmin" is the AI sales assistant persona. She is a warm, funny, Malaysian sales guide — a knowledgeable friend on WhatsApp, not a formal corporate representative.

5.1 Language Rules

- Detects the language of the customer's latest message and replies in the same language.
- Chinese message → full reply in Simplified Chinese with a natural Malaysian-Chinese tone.
- English message → English reply with light Malaysian flavour.
- Malay message → Malay reply with light Malaysian slang used sparingly.
- Must not switch languages mid-conversation.

5.2 Greeting Policy (Dynamic)

The system dynamically decides whether a greeting is allowed:

- ALLOW: the current message is only a greeting word AND there is no previous chat history.
- FORBID: the message contains an actual question/statement, or history already exists.

5.3 Strictly Forbidden Behaviours

- No formal self-introduction.
- No long company marketing paragraphs.
- No more than 2–3 short sentences per reply unless necessary.
- Never asks for the customer's name or phone number (already collected).
- Never answers if the knowledge base doesn't have the info — escalates instead.

5.4 Core Tone & Behaviour Guidelines

Guideline	Description
Warm & Welcoming	Responds as if genuinely happy the customer messaged
Helpful, not pushy	Guides and suggests, never pressures; no hard-sell tactics
Funny & relaxed	Light positive humour allowed; no sarcasm, no rude jokes
Local tone	Malaysian conversational touches used sparingly and naturally
Encouraging	Positive reinforcement on customer decisions
Emotionally aware	Acknowledges frustration or worry before helping

Conversational	Ends with one friendly question to keep the conversation flowing
One question at a time	Never combines multiple questions in one reply
WhatsApp style	1–2 short sentences; no long paragraphs; no robotic tone

5.5 Escalation Fallback Rule

When a test ride, trade-in, or service-booking topic arises (or an FAQ entry is flagged for escalation), Jasmin must send only one short, fixed fallback line — in the customer's language — indicating that a human team member will follow up shortly, with no additional content in the same reply.

6. Test Ride Booking System — Business Rules

6.1 Event Details

Parameter	Value
Event	Test-ride weekend event
Available Time Slots	10:00 / 11:00 / 12:00 / 14:00 / 15:00 / 16:00 (local time)
Duration Per Slot	1 hour
Sessions Per Slot	3 sessions (S1, S2, S3) running in parallel
Capacity Per Session	5 riders
Total Capacity Per Hour	15 riders (3 sessions × 5)
Max Bookings Per Rider	1 booking per rider for the entire event (1 model only)

6.2 Available Models (Category Overview)

The system supports test rides across several brands, each mapped to a license requirement (Full B for 500cc+, B2 for 250cc and below) and a brand-specific digital indemnity form. Exact model lists and form links are maintained in the internal configuration and are not reproduced here.

6.3 Compulsory Riding Gear

All riders must bring: helmet, gloves, riding jacket, and covered shoes. This is mentioned in every booking confirmation message.

7. Setup & Deployment Guide (Summary)

This section summarizes what a new developer needs to set up and deploy the workflow. Exact credential values, base/table identifiers, and phone number IDs are intentionally omitted — obtain these from the secure internal credentials store.

7.1 Prerequisites

- An n8n instance (self-hosted or cloud).
- An OpenRouter account with API access to the chosen LLM model.

- A WhatsApp-enabled messaging account (Cloud API or equivalent).
- A database (Airtable) with the required tables set up (see Section 2.4).
- A Google account with the Calendar API enabled (OAuth2).
- A WhatsApp Business Platform account with a configured phone number.

7.2 Database Setup

Create the tables described in Section 2.4 with the field names and types noted there (text, long text, checkbox, date/time fields as appropriate). Use a Status field on the staff table to mark inactive staff, so they are automatically excluded from escalation rotation.

7.3 Credential Setup (n8n)

- LLM API key: create a header-auth credential for the LLM provider (bearer token in the Authorization header).
- Database API access: create a personal access token credential scoped to read/write records and read schema.
- Calendar: create an OAuth2 credential authorized against the target calendar account; ensure the Calendar API is enabled in the cloud console.
- WhatsApp: create a WhatsApp Business Cloud API credential with the appropriate access token and phone number ID.

7.4 Importing the Workflow

- In n8n, go to Workflows → Import and upload the workflow JSON file.
- After import, re-link every node that uses credentials to the credentials created in your own instance (credential IDs will differ).
- Verify and update any base/table identifiers to match your own database setup.
- Activate the workflow.

7.5 Inbound Message Webhook Setup

Since the workflow uses a polling approach rather than a direct webhook into n8n, a separate integration must write incoming WhatsApp messages into the staging table with all required fields (message ID, From, To, Body, ProfileName, Processed = false, Timestamp). The main workflow then picks up new records on its next poll cycle.

7.6 WhatsApp Template Setup

The escalation notification uses a WhatsApp Business template that must be pre-approved by the messaging provider before use. Create a template with 5 variable placeholders, submit it for approval, and once approved reference its exact name and language in the "Send template" node.

8. Maintenance & Operational Guide

8.1 Adding / Removing Staff

- To add staff: create a new record with Name, phone-number ID, and Status = "Active."
- To remove staff: change Status to "Inactive" — the staff-lookup node filters out inactive staff automatically.
- Round-robin rotation follows the alphabetically sorted order of active staff.

8.2 Updating FAQ Content

- Add, edit, or delete FAQ rows directly in the knowledge-base table.
- Ensure the Category field matches one of the categories the LLM classifier can assign.
- Use the Remarks field to flag rows that require escalation.
- Changes take effect immediately on the next message-processing cycle — no deployment needed.

8.3 Changing the AI Model

Locate every "Prepare final LLM Payload" node (12 in total), update the model string in each, confirm the new model is available through your API account, and test with a few sample messages before full deployment.

8.4 Updating Booking Event Dates

- Open the deterministic booking-parser node and update the date-validation logic to match the new event dates.
- Update the date-parsing section for alternate date formats.
- Update the field-validation prompt that lists the allowed event dates.
- Update any hardcoded date references used when checking calendar slot availability.

8.5 Monitoring & Debugging

- All conversations are logged with full user message and bot reply text.
- Escalations are logged with staff assignment and timestamp.
- n8n's execution history shows every workflow run with node-by-node status.
- Failed nodes appear in red in n8n's execution view with error details.
- If messages pile up unprocessed, check that the schedule trigger is active and the workflow is enabled.

8.6 Common Issues & Solutions

Issue	Solution
Messages not being processed	Check the workflow is active, unprocessed records exist in the staging table, and the schedule trigger is running.
Bot replies in the wrong language	Check the prompt-building node — language detection follows the customer's most recent message.
LLM returning errors	Check API account credits/rate limits and that the LLM credential is still valid.
Calendar booking fails	Check that the calendar OAuth2 token hasn't expired; re-authorize if needed.
Staff not receiving escalation	Check the WhatsApp template is approved, staff phone number format is correct, and the phone number ID configuration is correct.
Duplicate bookings appearing	Ensure calendar event descriptions include the customer's phone number, which the duplicate-check logic relies on.
FAQ not found for a query	If no FAQ rows match the classified category, the bot falls back to its own general knowledge.

9. Node Inventory

The workflow contains 141 nodes across functional groups A–K, covering message ingestion, customer management, conversation history, intent classification, response generation, delivery/logging, escalation, follow-up automation, and the booking/reschedule/cancellation pipelines. A full per-node inventory (exact node names, types, and wiring) is maintained in the internal workflow JSON file itself, which should be treated as the definitive reference — the functional summaries in Section 3 describe what each group does.

10. Quick Reference Cheat Sheet

Item	Value
AI Model	Grok-4-Fast (via OpenRouter)
Polling Interval	Every 15 seconds
Follow-up Wait	1 hour after sending a reply
Max Chat History	Last 6 exchanges per customer
Max Riders Per Hour	15 (3 sessions × 5 riders)
Valid Booking Time Slots	10:00, 11:00, 12:00, 14:00, 15:00, 16:00 (13:00 is not a valid slot)

LLM Call Temperature Settings:

Use Case	Temperature	Reason
FAQ Category Classification	0.1	Highly deterministic — must pick an exact category
Main Sales Reply (Jasmin)	0.7	Creative, varied, human-sounding responses
Escalation Chat Summary	0.5	Balanced — accurate but readable

Redaction Notice

The following categories of information have been intentionally removed from this document: credential/API key values and credential IDs, database base/table identifiers, calendar account email address, WhatsApp phone number ID, and any other direct account identifiers. Please refer to the secure internal credentials store for this information.